

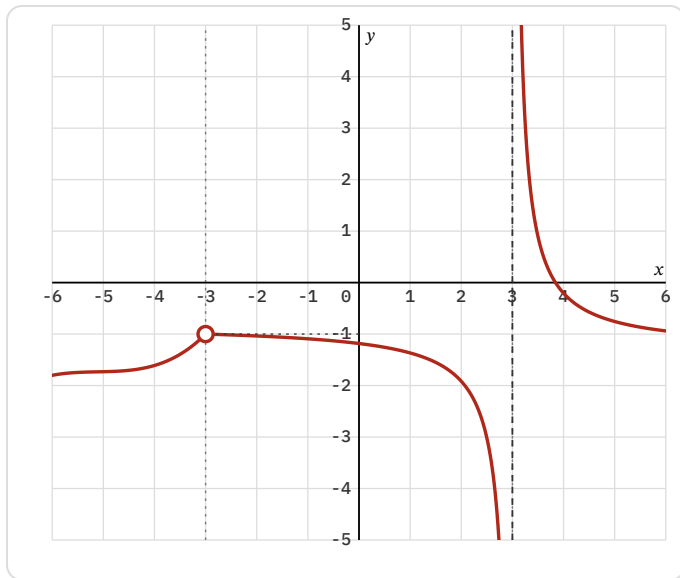
Mock Midterm A

Practice midterm (AI-Free) · Sections 1.1–1.4

Directions: Answer each question below. Be sure to show work to receive full credit. Work this exam on your own, without notes or AI, the way you will on the proctored midterm. Be sure the notation in your solutions matches the notation from the lesson videos. Write **DNE** if a value or limit does not exist. Suggested time: 75 minutes.

1.

Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write **DNE**.



a.) $f(-3)$ _____ (1)

b.) $\lim_{x \rightarrow -3^-} f(x)$ _____ (1)

c.) $\lim_{x \rightarrow -3^+} f(x)$ _____ (1)

d.) $\lim_{x \rightarrow -3} f(x)$ _____ (1)

e.) $f(3)$ _____ (1)

f.) $\lim_{x \rightarrow 3^-} f(x)$ _____ (1)

g.) $\lim_{x \rightarrow 3^+} f(x)$ _____ (1)

h.) $\lim_{x \rightarrow 3} f(x)$ _____ (1)

8 pts

2. Let $f(x) = \frac{x-4}{\sqrt{x}-2}$. Use the table to see what the outputs do as x approaches 4. 7 pts

x	3.5	3.9	3.99	4	4.01	4.1	4.5
$f(x)$	3.870829	3.974842	3.997498	?	4.002498	4.024846	4.12132

- a.) Can you find $\lim_{x \rightarrow 4} f(x)$ just by plugging in $x = 4$? (1)
 (A) Yes — $f(4)$ is defined, so substitute
 (B) No — $x = 4$ makes a denominator 0, so $f(4)$ is undefined
- b.) As $x \rightarrow 4^-$, $f(x)$ approaches... (2)
- c.) As $x \rightarrow 4^+$, $f(x)$ approaches... (2)
- d.) $\lim_{x \rightarrow 4} f(x) =$ (2)

3. Suppose 4 pts

$$\lim_{x \rightarrow 5} f(x) = 3, \quad \lim_{x \rightarrow 5} g(x) = 5, \quad \lim_{x \rightarrow 5} h(x) = 16$$

Use the properties of limits to find each limit.

- a.) $\lim_{x \rightarrow 5} x^2 f(x)$ (1)
- b.) $\lim_{x \rightarrow 5} \sqrt{h(x)} f(x)$ (1)
- c.) $\lim_{x \rightarrow 5} ([f(x)]^3 - g(x))$ (1)

- d.) $\lim_{x \rightarrow 5} [2f(x) + g(x)]^2$ (1)

4.

Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE.

a.) $\lim_{h \rightarrow 0} \frac{\frac{1}{3+h} - \frac{1}{3}}{h}$ _____ (2)

b.) $\lim_{x \rightarrow -2} \frac{3x^2 + 3x - 6}{2x^2 + 7x + 6}$ _____ (2)

4 pts

5.

Compute each limit algebraically. Use ∞ / $-\infty$ for unbounded results.

6 pts

a.) $\lim_{x \rightarrow -\infty} \frac{4x^2}{-x^4 - 7x^2 + 4x}$ _____ (2)

b.) $\lim_{x \rightarrow \infty} \frac{5x}{4x - 5}$ _____ (2)

c.) $\lim_{x \rightarrow \infty} \frac{4x^3 - 6x^2}{-4x^2}$ _____ (2)

6.

Let

7 pts

$$G(x) = \begin{cases} x^2 + 2 & x < 0 \\ 2x + 2 & x > 0 \end{cases}$$

Find each quantity. If it does not exist, write **DNE**.

a.) $G(0)$ _____ (1)

b.) $\lim_{x \rightarrow 0^-} G(x)$ _____ (1)

c.) $\lim_{x \rightarrow 0^+} G(x)$ _____ (1)

d.) $\lim_{x \rightarrow 0} G(x)$ _____ (1)

e.) $G(2)$ _____ (1)

f.) $\lim_{x \rightarrow -\infty} G(x)$ _____ (1)

g.) $\lim_{x \rightarrow \infty} G(x)$ _____ (1)

7.

Let

6 pts

$$f(x) = \begin{cases} \frac{2x^2 + 11x + 15}{2x + 5} & x < -\frac{5}{2} \\ 3x + 8 & x > -\frac{5}{2} \end{cases}$$

Find each limit or function value algebraically. Write **DNE** if it doesn't exist.

a.) $\lim_{x \rightarrow 1} f(x)$ _____ (1)

b.) $\lim_{x \rightarrow -5} f(x)$ _____ (1)

c.) $f\left(-\frac{5}{2}\right)$ _____ (1)

d.) $\lim_{x \rightarrow -\frac{5}{2}^-} f(x)$ _____ (1)

e.) $\lim_{x \rightarrow -\frac{5}{2}^+} f(x)$ _____ (1)

f.) $\lim_{x \rightarrow -\frac{5}{2}} f(x)$ _____ (1)

8.

Let

5 pts

$$f(x) = \begin{cases} -x^2 + 4x & x \leq 0 \\ -3x & x > 0 \end{cases}$$

Determine whether $f(x)$ is continuous at $x = 0$.

a.) $f(0)$ _____ (1)

b.) $\lim_{x \rightarrow 0^-} f(x)$ _____ (1)

c.) $\lim_{x \rightarrow 0^+} f(x)$ _____ (1)

d.) $\lim_{x \rightarrow 0} f(x)$ _____ (1)

e.) Is f continuous at $x = 0$? (1)

(A) Yes — $f(0)$ exists, the limit exists, and they are equal

(B) No — the one-sided limits differ, so the limit does not exist

(C) No — the limit exists but does not equal $f(0)$

(D) No — $f(0)$ is undefined

9.

Classify each function at the given x -value.

3 pts

a.) $f(x) = \begin{cases} x - 2 & x < -1 \\ x - 5 & x \geq -1 \end{cases}$ at $x = -1$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

b.) $f(x) = 3x^2 + x + 4$ at $x = 4$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

c.) $f(x) = \frac{x+2}{x^2-4}$ at $x = 2$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

10.

Let $s(t) = -16t^2 + 96t + 34$, where $s(t)$ represents the height, in feet, of a ball t seconds after it is thrown upward. Find the average rate of change over $[1, 3]$.

a.) $s(1)$ _____ (2)

b.) $s(3)$ _____ (2)

c.) Average rate of change (number only) _____ (2)

d.) Which sentence correctly states the result **with units**? (1)

(A) Between $t = 1$ and $t = 3$, the height of the ball changes at an average rate of **32** seconds per foot.

(B) Between $t = 1$ and $t = 3$ seconds, the height of the ball changes at an average rate of **32** feet per second.

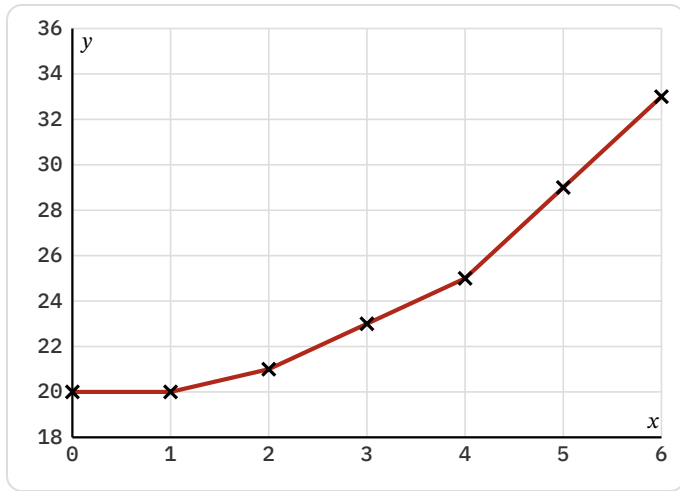
(C) At exactly $t = 3$ seconds, the height of the ball is changing at a rate of **32** feet per second.

(D) The height of the ball changes by a total of **32** feet from $t = 1$ to $t = 3$.

7 pts

11.

The graph shows the closing price (in dollars) of a stock at the start of each month in 2026, where $x = 0$ is January, $x = 1$ is February, and so on.



a.) Average rate of change on $[0, 4]$ (dollars per month) _____ (2)

b.) On which interval is the average rate of change greatest? (1)

(A) $[0, 2]$

(B) $[2, 4]$

(C) $[4, 6]$

3 pts

12. Let $f(x) = x^3$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table.
- a.) Simplified difference quotient (in terms of x and h) _____ (3)
- b.) $x = -1, h = 1$ _____ (1)
- c.) $x = -1, h = 0.1$ _____ (1)
- d.) $x = -1, h = 0.01$ _____ (1)
- e.) As $h \rightarrow 0$, what value do these approach? _____ (1)

7 pts

13.

Let $f(x) = -x^2 + x$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$.

a.) $f'(x) =$ _____ (3)

b.) $f'(-2) =$ _____ (2)

c.) Slope of the tangent line to f at $x = 2$ _____ (2)

d.) What does $f'(-2) = 5$ mean? (1)

(A) The slope of the secant line from $x = -2$ to $x = -2 + 1$ is 5.

(B) The instantaneous rate of change of f at $x = -2$ is 5 — the slope of the tangent line there.

(C) The average rate of change of f from $x = 0$ to $x = -2$ is 5.

(D) The value of f at $x = -2$ is 5.

8 pts

14. Let $f(x) = -2x^2 + 5x + 5$, so $f'(x) = -4x + 5$. 6 pts

a.) At what x is the tangent line horizontal? (2)

b.) What is the y -coordinate of that point? (2)

c.) At what x does the tangent line have slope -2 ? (2)

15.

Let $f(x) = x^2 - 3$. Complete the limit definition, then interpret $f'(-3)$.

6 pts

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{((x+h)^2 - 3) - (x^2 - 3)}{h} = ?$$

a.) $f'(x) =$ _____ (3)

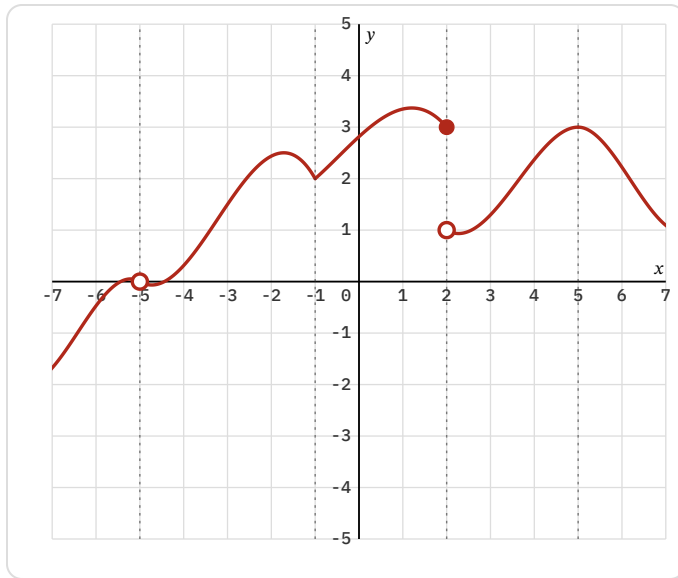
b.) $f'(-3)$ _____ (2)

c.) What does $f'(-3) = -6$ tell you? (1)

(A) The slope of the tangent line to the graph of f at $x = -3$ is -6.(B) The slope of the secant line from $x = 0$ to $x = -3$ is -6.(C) The point $(-3, -6)$ is on the graph of f .(D) The graph of f crosses the y -axis at -6.

16.

At each labeled x -value, decide whether the graph of f is continuous there and whether it is differentiable there.



a.) At $x = -5$ (1)

- (A) Continuous and differentiable
- (B) Continuous but NOT differentiable
- (C) Not continuous (so not differentiable)

b.) At $x = -1$ (1)

- (A) Continuous and differentiable
- (B) Continuous but NOT differentiable
- (C) Not continuous (so not differentiable)

c.) At $x = 2$ (1)

- (A) Continuous and differentiable
- (B) Continuous but NOT differentiable
- (C) Not continuous (so not differentiable)

d.) At $x = 5$ (1)

- (A) Continuous and differentiable
- (B) Continuous but NOT differentiable
- (C) Not continuous (so not differentiable)

4 pts

